DIMENSION THEORY FOR LOCALLY ALGEBRAIC K-SCHEMES: CATENARITY, DIMENSION AT A POINT, AND TRANSCENDENCE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's opening paragraph explicitly announces the strategy of reducing to finite type k-algebras and citing the commutative algebra chapter; that paragraph is the first block of the proof and is reproduced as written.

Editorial note. The counted spans carry 35 navigation stubs, one for each labelled item invoked outside the span; none of them replaces a load-bearing step, since the reductions themselves are written out inside the span. No bibliography citation occurs in the spans.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-dimension-locally-algebraic is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Catenary topological space. Topology, Definition topology-definition-catenary, named in the statement at line 3789. Supplies the notion verified in part (1).
- Finitely generated k-algebras of finite type: catenarity, dimension at a point, dimension of polynomial rings. Algebra, Lemmas algebra-lemma-catenary, algebra-lemma-quotient-catenary, algebra-lemma-dimension-height-polynomial-ring, algebra-lemma-dimension-closed-point-finite-type-field, algebra-lemma-dimension-at-a-point-finite-type-field, algebra-lemma-dimension-at-a-point-finite-type-over-field, algebra-lemma-dimension-spell-it-out, algebra-lemma-dimension-prime-polynomial-ring. Supplies the affine case to which every part of the statement is reduced.
- Normality criterion as the intersection of the localizations at height one primes. Algebra, Lemma algebra-lemma-normal-domain-intersection-localizations-height-1, used elsewhere in the chapter and cited in the sibling lemma; here the affine reductions rest on the corresponding algebra inputs. Supplies that affine pieces can be studied through their height one localizations.
- Jacobson property of schemes locally of finite type over a field. Same chapter, Lemma lemma-locally-finite-type-Jacobson, invoked at proof line 3879. Supplies the closed point inside a nonempty open subset used in part (3).

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00NM: sources/stacks-chow/tags.txt, source lines 864–864. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02IH: sources/stacks-chow/tags.txt, source lines 3300–3300. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00P1: sources/stacks-chow/tags.txt, source lines 915–915. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00OT: sources/stacks-chow/tags.txt, source lines 907–907. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00OU: sources/stacks-chow/tags.txt, source lines 908–908. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 00P0: sources/stacks-chow/tags.txt, source lines 914–914. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00OS: sources/stacks-chow/tags.txt, source lines 906–906. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00NK: sources/stacks-chow/tags.txt, source lines 862–862. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0478: sources/stacks-varieties/pinned-original-source.tex, source lines 2670–2670. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 02I1: sources/stacks-chow/tags.txt, source lines 3284–3284. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 0058: sources/stacks-chow/tags.txt, source lines 200–200. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02I2: sources/stacks-chow/tags.txt, source lines 3285–3285. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 004W: sources/stacks-chow/tags.txt, source lines 188–188. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme.*

(1) *The topological space of X is catenary (Topology, Definition 02I1).*
(2) *For $x \in X$ closed, we have $\dim_x(X) = \dim(\mathcal{O}_{X,x})$.*
(3) *For X irreducible we have $\dim(X) = \dim(U)$ for any nonempty open $U \subset X$ and $\dim(X) = \dim_x(X)$ for any $x \in X$.*
(4) *For X irreducible any chain of irreducible closed subsets can be extended to a maximal chain and all maximal chains of irreducible closed subsets have length equal to $\dim(X)$.*
(5) *For $x \in X$ we have $\dim_x(X) = \max \dim(Z) = \min \dim(\mathcal{O}_{X,x'})$ where the maximum is over irreducible components $Z \subset X$ containing x and the minimum is over specializations $x \rightsquigarrow x'$ with x' closed in X.*
(6) *If X is irreducible with generic point x, then $\dim(X) = trdeg_k(\kappa(x))$.*
(7) *If $x \rightsquigarrow x'$ is an immediate specialization of points of X, then we have $trdeg_k(\kappa(x)) = trdeg_k(\kappa(x')) + 1$.*
(8) *The dimension of X is the supremum of the numbers $trdeg_k(\kappa(x))$ where x runs over the generic points of the irreducible components of X.*
(9) *If $x \rightsquigarrow x'$ is a nontrivial specialization of points of X, then*
   (a) $\dim_x(X) \leq \dim_{x'}(X)$,
   (b) $\dim(\mathcal{O}_{X,x}) < \dim(\mathcal{O}_{X,x'})$,
   (c) *$trdeg_k(\kappa(x)) > trdeg_k(\kappa(x'))$, and*
   (d) *any maximal chain of nontrivial specializations $x = x_0 \rightsquigarrow x_1 \rightsquigarrow \ldots \rightsquigarrow x_n = x'$ has length $n = trdeg_k(\kappa(x)) - trdeg_k(\kappa(x'))$.*
(10) *For $x \in X$ we have $\dim_x(X) = trdeg_k(\kappa(x)) + \dim(\mathcal{O}_{X,x})$.*
(11) *If $x \rightsquigarrow x'$ is an immediate specialization of points of X and X is irreducible or equidimensional, then $\dim(\mathcal{O}_{X,x'}) = \dim(\mathcal{O}_{X,x}) + 1$.*

Proof. Instead on relying on the more general results proved earlier we will reduce the statements to the corresponding statements for finite type k-algebras and cite results from the chapter on commutative algebra.

Proof of (1). This is local on X by Topology, Lemma 02I2. Thus we may assume $X = \mathrm{Spec}(A)$ where A is a finite type k-algebra. We have to show that A is catenary (Algebra, Lemma 02IH). We can reduce to $k[x_1, \ldots, x_n]$ using Algebra, Lemma 00NK and then apply Algebra, Lemma 00OR. Alternatively, this holds because k is Cohen-Macaulay (trivially) and Cohen-Macaulay rings are universally catenary (Algebra, Lemma 00NM).

Proof of (2). Choose an affine neighbourhood $U = \mathrm{Spec}(A)$ of x. Then $\dim_x(X) = \dim_x(U)$. Hence we reduce to the affine case, which is Algebra, Lemma 00OU.

Proof of (3). It suffices to show that any two nonempty affine opens $U, U' \subset X$ have the same dimension (any finite chain of irreducible subsets meets an affine open). Pick a closed point x of X with $x \in U \cap U'$. This is possible because X is irreducible, hence $U \cap U'$ is nonempty, hence there is such a closed point because X is Jacobson by Lemma 0478. Then $\dim(U) = \dim(\mathcal{O}_{X,x}) = \dim(U')$ by

Algebra, Lemma 00OS (strictly speaking you have to replace X by its reduction before applying the lemma).

Proof of (4). Given a chain of irreducible closed subsets we can find an affine open $U \subset X$ which meets the smallest one. Thus the statement follows from Algebra, Lemma 00OS and $\dim(U) = \dim(X)$ which we have seen in (3).

Proof of (5). Choose an affine neighbourhood $U = \mathrm{Spec}(A)$ of x. Then $\dim_x(X) = \dim_x(U)$. The rule $Z \mapsto Z \cap U$ is a bijection between irreducible components of X passing through x and irreducible components of U passing through x. Also, $\dim(Z \cap U) = \dim(Z)$ for such Z by (3). Hence the statement follows from Algebra, Lemma 00OT.

Proof of (6). By (3) this reduces to the case where $X = \mathrm{Spec}(A)$ is affine. In this case it follows from Algebra, Lemma 00P0 applied to A_{red}.

Proof of (7). Let $Z = \overline{\{x\}} \supset Z' = \overline{\{x'\}}$. Then it follows from (4) that $Z \supset Z'$ is the start of a maximal chain of irreducible closed subschemes in Z and consequently $\dim(Z) = \dim(Z') + 1$. We conclude by (6).

Proof of (8). A simple topological argument shows that $\dim(X) = \sup \dim(Z)$ where the supremum is over the irreducible components of X (hint: use Topology, Lemma 004W). Thus this follows from (6).

Proof of (9). Part (a) follows from the fact that any open $U \subset X$ containing x' also contains x. Part (b) follows because $\mathcal{O}_{X,x}$ is a localization of $\mathcal{O}_{X,x'}$ hence any chain of primes in $\mathcal{O}_{X,x}$ corresponds to a chain of primes in $\mathcal{O}_{X,x'}$ which can be extended by adding $\mathfrak{m}_{x'}$ at the end. Both (c) and (d) follow formally from (7).

Proof of (10). Choose an affine neighbourhood $U = \mathrm{Spec}(A)$ of x. Then $\dim_x(X) = \dim_x(U)$. Hence we reduce to the affine case, which is Algebra, Lemma 00P1.

Proof of (11). If X is equidimensional (Topology, Definition 0058) then $\dim(X)$ is equal to the dimension of every irreducible component of X, whence $\dim_x(X) = \dim(X) = \dim_{x'}(X)$ by (5). Thus this follows from (7). $\square$